

Question 1 Basics (6 Marks)

5.5

- (1) We want to create a virtual book of a novel with turning pages (Figure 1). What is the method to animate the turning page? What are the parameters to control in the method? How the parameters will be controlled? Assume that we do not need to consider the paper deformation during the turning.



Figure 1. A virtual book with turning pages

Answer:

Method: Keyframing.

parameter to control: t (time)

How to control: by interpolating points.

Given key positions (x_i, y_i, t_i) , $i=1, 2, \dots, n$
we use interpolation to find the curve

$$C(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \text{ s.t. } C(t_i) = \begin{bmatrix} x_i \\ y_i \end{bmatrix}$$

- (2) We want to animate a falling ball (Figure 2). What is the method to animate? What are the parameters to control in the method? How the parameters will be controlled? Assume that we do not need to consider the ball deformation during the falling.

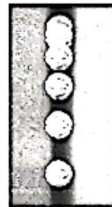


Figure 2. A falling ball

Answer:

Method: Keyframing

parameter: t (time)

how to control: linear interpolation.

physical
height

Question 2 Project (6 Marks)

5-5

(1) Please describe briefly the following three components in computer animation pipeline:

Answer:

Storyboard: typically are paper and pencil sketches on individual sheets taped on a wall, and specifies the key ~~scenes~~ scenes, camera moves and edits, and character gross ~~and~~ motion.

Modeling: The process to model a character out, specifying how the character will be looked like

Rigging:

Animation: The process to create continuous frames between 2 keyframes with lighting, ~~camera~~ and rendering.
 and recorded by cameras
 not only keyframes

(2) In the assignment 1 "Short Animation" project, what is your major contribution? What are the functions in which software are you using?

Answer:

In the Keyframes ~~writi~~ drawing stage, I discussed with my teammates to generate some ideas, also ~~based~~ wrote ~~ex~~ explanations and descriptions corresponding to each keyframes drawn by Sitlan and Alicia. (in google doc)

In the keyframe rendering stage, I use Blender to render the 1st scene.

Question 3 3D Rotation (8 marks)

- (1) Given a 3D rotation around an axis $(1, 1, 1)^T$ with 90 degrees, what are the unit quaternions representing the same rotation? Derive the result of this rotation on the point $(1, 1, 0)^T$ using the unit quaternions.

Let $d = (\cos 45^\circ, \sin 45^\circ (1, 1, 1)) = (\frac{\sqrt{2}}{2}, (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}))$
 Answer: ~~$P = (1, 1, 0)^T$. Let $P = (0, \underline{p}) = (0, (1, 1, 0))$
 unit quaternion: $q = \frac{d}{\|d\|} = (\frac{\sqrt{2}}{2}, (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}))$
 Let $\underline{p} = (1, 1, 0)^T$. $P = (0, \underline{p}) = (0, (1, 1, 0))$~~

On the back

- (2) Given the two rotations of the coordinate system OXYZ: First, rotate around Z-axis 45 degrees getting the new coordinate system O'X'Y'Z'. Second, rotate around the new X'-axis 90 degrees. If we want to achieve the rotation of OXYZ in one-step, derive the axis and angle of the rotation.

Answer:

For rotation 1:

$$\theta = 45^\circ \quad \underline{v}_1 = (0, 0, 1)^T$$

$$q_1 = (\cos 22.5^\circ, \sin 22.5^\circ (0, 0, 1))$$

For rotation 2:

$$\theta = 90^\circ, \quad \underline{v}_2 = (\cos 45^\circ, \sin 45^\circ (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0))$$

$$\underline{v}_2 = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T$$

$$q_2 = (\cos 45^\circ, \sin 45^\circ (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0))$$

Thus, the axis is $(0.71, 0.71, 0)$

and the angle is 117.59° .

back

$$\therefore q = q_2 \circ q_1$$

$$= (\frac{\sqrt{2}}{2}, (\frac{1}{2}, \frac{1}{2}, 0)) \circ (\cos \frac{\pi}{8}, (0, 0, \sin \frac{\pi}{8}))$$

$$= (\frac{\sqrt{2}}{2} \cos \frac{\pi}{8} - 0, (0, 0, \frac{\sqrt{2}}{2} \sin \frac{\pi}{8}) + (\frac{\cos \frac{\pi}{8}}{2}, \frac{\cos \frac{\pi}{8}}{2}, 0) + (\frac{\sin \frac{\pi}{8}}{2}, \frac{\sin \frac{\pi}{8}}{2}, 0))$$

$$= (\frac{\sqrt{2} \cdot \cos \frac{\pi}{8}}{2}, (\frac{\cos \frac{\pi}{8} + \sin \frac{\pi}{8}}{2}, \frac{\cos \frac{\pi}{8} + \sin \frac{\pi}{8}}{2}, \frac{\sqrt{2} \sin \frac{\pi}{8}}{2})) = (0.65, 0.65, 0.65, 0.65)$$

$$\hat{=} (117.59^\circ, (117.59, 117.59, 48.71))$$

Q 3.

(1) Let $\theta = 90^\circ$, $\underline{u} = (1, 1, 1)^T$

$\therefore d = (\cos \frac{\theta}{2}, \sin \frac{\theta}{2} \underline{u})$

$\therefore d = (\cos 45^\circ, \sin 45^\circ (1, 1, 1))$

$= (\frac{\sqrt{2}}{2}, (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}))$

need to normalize first

$\therefore$ unit quaternion: $q = \frac{d}{\|d\|} = (\frac{1}{2}, (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}))$

$\therefore$ conjugation: $q^* = (\frac{1}{2}, (-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}))$

Let $\underline{p} = (1, 1, 0)^T$

Let $P = (0, \underline{p}) = (0, (1, 1, 0))$

Let $\underline{p}' = (x, y, z)$ is the point coordinates after rotation

$\therefore (0, (x, y, z)) = q \circ P \circ q^*$

$= (\frac{1}{2}, (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})) \circ (0, (1, 1, 0)) \circ (\frac{1}{2}, (-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}))$

$= (-1, (0, 1, 0)) \circ (\frac{1}{2}, (-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}))$

$= (0, (0, 1, 1))$

Thus, the unit quaternion is $(\frac{1}{2}, (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}))$,

and the resultant point is $(0, 1, 1)$.

(2) After:

unit quaternion: $(0.56, (0.56, 0.56, 0.23))$

Thus, the angle is $\cos^{-1}(0.56) \times 2 =$

axis?